Photo Shooting Study: Data Analysis Plan

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This is the data-analysis part of the preregistration: hypotheses, variables, analyses, predicted results, exclusions, sample size and further analyses.

Hypotheses

Does the amount of control a person has while taking a photo affect how well they later recognise the photographed picture?

- **H1 (manipulation check).** Felt control is highest under full control, lower under medium and lowest under low control (F > M > L).
- **H2 (main hypothesis).** Pictures photographed under more control are recognised more often in a surprise memory test (F > M > L).
- **H3 (recognition speed).** Pictures photographed under more control are recognised faster: correct “old” responses are fastest after full and slowest after low control (F < M < L in reaction time).

Variables

- **Felt control:** rating after every photo, “How much control did you feel over the frame's movement?”, on a slider from 1 (no control) to 7 (full control).
- **Recognition (main DV):** for each of the 180 photographed (old) pictures, whether the participant calls it “old” in the recognition test (hit = 1) or “new” (miss = 0).
- **Recognition reaction time:** time from picture onset to the old/new key press, in seconds, analysed for hits only.
- **False alarms:** for each of the 180 new pictures, whether the participant wrongly calls it “old”.

- **Sensitivity d′ (secondary):** per participant and condition, d′ = z(hit rate in that condition) − z(false-alarm rate), with rates corrected as (count + 0.5) / (number of trials + 1).

Where each variable comes from in the data file (one CSV per participant, one row per trial):

Variable	Rows	Columns
Condition (F / M / L)	encoding trials (`phase == "test"`)	`condition`
Felt control	encoding trials	`control_rating`
Hit	recognition trials (`phase == "recognition"`) with `ground_truth == "seen"`	`response == "yes"` ; condition from `encoding_condition`
Hit reaction time	the hits above	`rt`
False alarm	recognition trials with `ground_truth == "unseen"`	`response == "yes"` ; `encoding_condition` is empty

Read `condition` and `encoding_condition` as text: R otherwise turns the code `F` into FALSE.

Analyses

All analyses are run in R with lme4. Condition is a factor with Full as the reference level. α = .05 for every test; pairwise comparisons within a hypothesis are Holm-corrected.

- **H1.** Linear mixed model on the 180 ratings per person: `rating ~ condition + (1 + condition | participant)` . F-test of condition (Satterthwaite degrees of freedom), followed by pairwise comparisons F–M, F–L and M–L.
- **H2.** Logistic mixed model on the 180 old pictures per person: `hit ~ condition + (1 + condition | participant) + (1 | picture)` . The test of H2 is a likelihood-ratio test comparing this model with the same model without condition. If it is significant, we report pairwise comparisons F–M, F–L and M–L as odds ratios with 95% confidence intervals and model-predicted hit probabilities per condition. H2 is supported if the likelihood-ratio test is significant and recognition is higher under full than under low control (odds ratio F vs L > 1). The test is two-sided. Full versus Low is the decisive comparison because it is the linear trend across the three ordered control levels (contrast weights −1, 0, +1) and the largest difference the hypothesis predicts; the adjacent comparisons F–M and M–L each carry about half the effect and a quarter of the power, so they describe the shape of the effect rather than decide H2.

- **H3.** Linear mixed model on the reaction times of hits (old pictures correctly called "old"): `log(rt) ~ condition + (1 + condition | participant) + (1 | picture)`. Reaction times are log-transformed because they are right-skewed. F-test of condition (Satterthwaite degrees of freedom), followed by pairwise comparisons F–M, F–L and M–L. Estimates are back-transformed and reported as typical reaction times per condition in seconds and as ratios between conditions (for example 0.95 = 5% faster). H4 is supported if the F-test is significant and hits are faster after full than after low control.

If a model does not converge or is singular, the random part is simplified in this order until it fits, and the final model is reported: (1) remove the correlations between random effects; (2) remove the random slope; (3) remove the random intercept for pictures. The random intercept for participants is never removed. Model assumptions are checked with simulated residuals (DHARMa) for the logistic models and residual plots for the linear models (H1, H4); if the H1 residuals are clearly non-normal, an ordinal mixed model is reported instead.

Packages. lme4 (mixed models), lmerTest (Satterthwaite F-tests), emmeans (pairwise comparisons), DHARMa (residual checks), afex (repeated-measures ANOVA for the secondary analysis).

Predicted results

We expect felt-control ratings to fall from full to medium to low control. We expect recognition of photographed pictures to be highest after full control and lowest after low control, with medium in between; the clearest difference should be full versus low control. We expect that on trials where a person felt more control than usual, the photographed picture is more likely to be recognised, even when the condition of the trial is taken into account. We expect correctly recognised pictures to be recognised faster after full than after low control, with medium in between.

Outliers and exclusions

Exclusions are applied before any analysis, in this order:

1. **Participants** are excluded if any of the following holds:

- the experiment was not completed;
- overall d′ ⩽ 0 (hit rate not above false-alarm rate), i.e. recognition at chance;
- the calibrated control level for Medium is not higher than for Low.

2. **Trials** are excluded if any of the following holds:
   - recognition trials with reaction times exceeding the participant's mean by more than 3 SD.

Excluded participants are replaced until the planned sample size is reached. No other data are removed.

Sample size and power

Effect size from the literature. Studies that compare an action with a strong sense of agency against one with a weak or no sense of agency report paired effect sizes between d = 0.34 and 0.60 for recognition memory: Hon & Yeo (2021) d = 0.48 and 0.42 (congruent vs incongruent), 0.34 and 0.52 (strong vs weak agency); Zhu et al. (2026) d ≈ 0.45, 0.59 and 0.43 in Experiments 1, 3 and 4 (converted from the reported F-tests). The largest study, Schreiner et al. (2024, N = 118), found d = 0.18 for d′, and Tsuji & Imaizumi (2022, N = 28) found no effect. Our manipulation varies the degree of control instead of agency versus no agency, so we plan for an effect somewhat below the typical published value: dz = 0.4 for Full versus Low control, with Medium halfway between.

Power simulation. Because the planned test is a mixed model, power was estimated by simulation in R 4.3.3, with own code that uses the packages `lme4` (1.1.35) to fit the logistic mixed model and `MASS` (7.3-60) to draw the random effects. **[Attach the simulation script.]** Each simulated data set has 60 old pictures per condition, a hit rate of .75 under full control, differences between people in overall memory (SD 0.7 on the logit scale), differences between pictures in memorability (SD 0.6) and differences between people in the size of the control effect (SD 0.25). For each sample size and effect size, 1,000 data sets were analysed with the test of condition at α = .05. A check with the preregistered logistic mixed model (200 data sets, N = 40, dz = 0.4) gave a power of .57, close to the .64 of the faster by-person test used for the grid.

N	dz = 0.3	dz = 0.4	dz = 0.5	Smallest dz detectable with 80% power
30	.28	.49	.71	0.59
40	.38	.64	.85	0.48
50	.42	.75	.92	0.43
60	.50	.82	.95	0.39

N	dz = 0.3	dz = 0.4	dz = 0.5	Smallest dz detectable with 80% power
80	.64	.93	.99	0.34
100	.75	.97	1.00	–

Decision. We will collect data until **[60]** participants with complete data that pass all exclusion criteria are reached, or until **[date]**, whichever comes first. With 60 participants the study has 82% power to detect dz = 0.4 for Full versus Low control and 95% power for dz = 0.5; the smallest effect detectable with 80% power is dz = 0.39. **[If 60 is not feasible in the lab-rotation period: with 40 participants, power is 64% for dz = 0.4 and the smallest effect detectable with 80% power is dz = 0.48; state the number you can reach and the power that goes with it.]** The adjacent comparisons Full–Medium and Medium–Low are each expected to be half the size of Full–Low and have power below .30 at every sample size in the table; they are reported descriptively.